Week 02 teacher guide

Ratios and matching amounts

Keep this guide beside PowerPoint slides 67-127. Follow the numbered slide sequence, then use the matching answer section after the student has tried.

The seven days

Day 008. Saturday | pages 36-42 | slides 67-84
Equivalent ratios

Day 009. Sunday | pages 43-52 | slides 85-103
Ratio tables and comparisons

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Ratios and matching amounts

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Weekly check

This week’s idea

A ratio compares two amounts in a chosen order. Equivalent ratios scale both amounts together. Tables and double number lines keep matching amounts connected.

Use the built-in Booster review where it appears in the day’s work. It replaces ordinary practice. No separate Booster packet is needed.

PDF bookmarks jump to each lesson and answer section. The student workbook’s Sunday reference pages are for checking an example after an attempt. Keep this teacher guide and the answer-review slides out of view during independent work.

Saturday. Equivalent ratios

Day 008, pages 36-42 | PowerPoint slides 67-84

Goal: I can keep a ratio unchanged while making more or less of a mixture.

Set out: two colors of counters, paper strips, pencil.

Check first: Multiplication, division, and equal groups.

Pacing: Retrieval 7; launch 8; explore 20; discuss 15; synthesis 7; exit 8; preview 5; slack 5. Total 75 min.

Explore: what to watch for

Says four red for every three blue. Why it matters: Names both quantities and their order. Tutor move: Ask for the reversed ratio and compare meanings.

Repeats or partitions a complete group. Why it matters: Shows equivalent ratios concretely. Tutor move: Ask which factor acts on both amounts.

Uses a half-batch or a unit batch. Why it matters: Non-whole factors can preserve the ratio. Tutor move: Accept either method if both quantities scale equally.

Likely misconceptions

Adds the same count to both colors. Why: Confuses additive and multiplicative change. Tutor move: Build identical batches.

Says 4 : 3 means four out of three. Why: Confuses part-to-part and part-to-whole. Tutor move: Count the total seven tiles before naming the red fraction.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 67. Week 2, Saturday

Topic: Equivalent ratios
Student: Watch and discuss

Say: I can keep a ratio unchanged while making more or less of a mixture.

Ask: What do you already know about this idea?

Expect: Multiplication, division, and equal groups.

If stuck: Ask for the reversed ratio and compare meanings.

Slide 68. Today

Topic: Equivalent ratios
Student: Watch and discuss

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: Equivalent ratios multiply or divide both quantities by the same factor.

If stuck: Read the learning goal together.

Slide 69. Recall | 4 min

Topic: Equivalent ratios

Student: Day 008 · Workbook p. 36 · R1, R2, R3, R4

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: $\frac{1}{6}$ liter; 2: $\frac{5}{8} > \frac{3}{5}$; 3: $\frac{1}{3} > \frac{1}{4}$; 4: $\frac{3}{5}$

If stuck: Give time to draw before giving a hint.

Slide 70. Recall: answers | 3 min

Topic: Equivalent ratios

Student: Day 008 · Workbook p. 36 · R1, R2, R3, R4

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: $\frac{1}{6}$ liter; $\frac{5}{8} > \frac{3}{5}$; $\frac{1}{3} > \frac{1}{4}$; $\frac{3}{5}$

If stuck: Rebuild one missed item with its model.

Slide 71. Read a picture and preserve order | 8 min

Topic: Equivalent ratios

Student: Day 008 · Workbook p. 37 · P1

Say: Read P1. Give time to try before comparing methods.

Ask: Which two quantities are paired? How can you keep the same relationship?

Expect: P1: 4 : 3; 3 : 4

If stuck: Put the quantities in two labeled rows. Find one matching pair before scaling.

Slide 72. Scale up by a whole number | 4 min

Topic: Equivalent ratios

Student: Day 008 · Workbook p. 37 · P2, P3

Say: Read P2 and P3. Give time to try before comparing methods.

Ask: Which two quantities are paired? How can you keep the same relationship?

Expect: P2: 9 nuts; P3: 4 : 3

If stuck: Put the quantities in two labeled rows. Find one matching pair before scaling.

Slide 73. The model | 4 min

Topic: Equivalent ratios

Student: Watch and discuss

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: Repeat the whole 4 : 3 group.

If stuck: Ask for the reversed ratio and compare meanings.

Slide 74. Non-whole scale factor | 3 min

Topic: Equivalent ratios
Student: Day 008 · Workbook p. 38 · P4, P5

Say: Read P4 and P5. Give time to try before comparing methods.

Ask: Which two quantities are paired? How can you keep the same relationship?

Expect: P4: 6 cups; P5: 6 cups yellow

If stuck: Put the quantities in two labeled rows. Find one matching pair before scaling.

Slide 75. Missing value in groups | 3 min

Topic: Equivalent ratios
Student: Day 008 · Workbook p. 38 · P6

Say: Read P6. Give time to try before comparing methods.

Ask: Which two quantities are paired? How can you keep the same relationship?

Expect: P6: 12 builders

If stuck: Put the quantities in two labeled rows. Find one matching pair before scaling.

Slide 76. Watch the thinking | 3 min

Topic: Equivalent ratios
Student: Watch and discuss · worked example

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 8 cups seeds

If stuck: Return to the matching unit in the model.

Slide 77. Your turn | 3 min

Topic: Equivalent ratios
Student: Day 008 · Workbook p. 39 · F

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 2 batches; 4 cups seeds

If stuck: Compare with the previous example without copying its numbers.

Slide 78. Two ways to see it | 8 min

Topic: Equivalent ratios

Student: Watch and discuss · compare the same example

Say: Partner compare: show the same problem in two ways.

Ask: How does each part of the model match a step in the calculation? Agree or disagree: Adding the same number to both counts keeps a mixture the same. Partner compare: How do you know the recipe with nine cups of oats has the same taste?

Expect: Accept drawings and equations that preserve the same quantities and units. Repeats or partitions a complete group.

If stuck: Ask which factor acts on both amounts.

Slide 79. A claim to check | 7 min

Topic: Equivalent ratios

Student: Day 008 · Workbook p. 39 · D

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: No. Two copies would give 8 : 6, not 7 : 6.

If stuck: Build two full copies of the original group and compare.

Slide 80. Words and meaning | 4 min

Topic: Equivalent ratios

Student: Watch and discuss

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: 4 red to 3 blue is 4 : 3. 4 : 3 and 8 : 6 are equivalent. From 4 : 3 to 12 : 9, the factor is 3.

If stuck: Read one definition and draw its example.

Slide 81. In your own words | 3 min

Topic: Equivalent ratios

Student: Day 008 · Workbook p. 36 · S

Say: Use your own words and one example.

Ask: Agree or disagree: Adding the same number to both counts keeps a mixture the same.

Expect: Equivalent ratios multiply or divide both quantities by the same factor.

If stuck: Point to the model and name what each part represents.

Slide 82. Exit ticket | 8 min

Topic: Equivalent ratios
Student: Day 008 · Workbook p. 40 · E

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: 21 green beads

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 83. The week ahead | 5 min

Topic: Equivalent ratios
Student: Days 010-014 · Workbook pp. 53-62

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 84. Tutor reference

Topic: Equivalent ratios
Student: Tutor only · hidden in slide show

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: 21 green beads

Got it: Gives 21 green beads and shows a factor of 3 on both quantities.

Almost there: Gives 21 with incomplete labeling, or makes a small arithmetic slip after using factor 3.

Needs re-teaching: Gives 15 or another answer from adding the same amount.

Sunday. Ratio tables and comparisons

Day 009, pages 43-52 | PowerPoint slides 85-103

Goal: I can use tables, double number lines, and equal totals to solve ratio problems.

Set out: grid paper, ruler, two colors of counters.

Check first: Scaling both amounts in a ratio.

Pacing: Retrieval 5; launch 5; explore 15; discuss 12; synthesis 5; mixed review 10; exit 8; preview 5. Total 65 min.

Explore: what to watch for

Keeps matching values in the same row. Why it matters: Pairs are meaningful together. Tutor move: Ask what one complete row means.

Finds a common water amount or compares part fractions. Why it matters: Both are valid ratio comparisons. Tutor move: Accept either method and require a shared comparison unit.

Adds ratio parts before using the total. Why it matters: Separates a part-to-part ratio from the whole. Tutor move: Ask the student to label all five parts.

Likely misconceptions

Compares mixtures by subtraction. Why: A difference ignores mixture size. Tutor move: Use the same amount of water in both mixtures.

Divides total by one ratio number. Why: Treats one part as the whole. Tutor move: Draw and count all ratio parts first.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 85. Week 2, Sunday

Topic: Ratio tables and comparisons
Student: Watch and discuss

Say: I can use tables, double number lines, and equal totals to solve ratio problems.

Ask: What do you already know about this idea?

Expect: Scaling both amounts in a ratio.

If stuck: Ask what one complete row means.

Slide 86. Today

Topic: Ratio tables and comparisons
Student: Watch and discuss

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: A ratio table records equal groups. Comparing mixtures needs a common amount or a common total.

If stuck: Read the learning goal together.

Slide 87. Recall | 3 min

Topic: Ratio tables and comparisons
Student: Day 009 · Workbook p. 43 · R1, R2, R3, R4

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: At 1 $\frac{3}{4}$, the seventh quarter-step from 0; 2: $\frac{5}{12}$; 3: Smaller; 6; 4: 2

If stuck: Give time to draw before giving a hint.

Slide 88. Recall: answers | 2 min

Topic: Ratio tables and comparisons
Student: Day 009 · Workbook p. 43 · R1, R2, R3, R4

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: At 1 $\frac{3}{4}$, the seventh quarter-step from 0; $\frac{5}{12}$; Smaller; 6; 2

If stuck: Rebuild one missed item with its model.

Slide 89. One missing table value | 5 min

Topic: Ratio tables and comparisons
Student: Day 009 · Workbook p. 44 · P1

Say: Read P1. Give time to try before comparing methods.

Ask: Which two quantities are paired? How can you keep the same relationship?

Expect: P1: 10 cups water

If stuck: Put the quantities in two labeled rows. Find one matching pair before scaling.

Slide 90. Solve the same problem two ways | 3 min

Topic: Ratio tables and comparisons
Student: Day 009 · Workbook p. 44 · P2

Say: Read P2. Give time to try before comparing methods.

Ask: Which two quantities are paired? How can you keep the same relationship?

Expect: P2: 25 cups water

If stuck: Put the quantities in two labeled rows. Find one matching pair before scaling.

Slide 91. The model | 3 min

Topic: Ratio tables and comparisons
Student: Watch and discuss

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: Two cups concentrate pairs with five cups water.

If stuck: Ask what one complete row means.

Slide 92. Compare using a common amount | 3 min

Topic: Ratio tables and comparisons
Student: Day 009 · Workbook pp. 44, 45 · P3, P4

Say: Read P3 and P4. Give time to try before comparing methods.

Ask: Which two quantities are paired? How can you keep the same relationship?

Expect: P3: Mix A; P4: 14 cups yellow and 21 cups blue

If stuck: Put the quantities in two labeled rows. Find one matching pair before scaling.

Slide 93. Ratio points in the first quadrant | 2 min

Topic: Ratio tables and comparisons
Student: Day 009 · Workbook p. 45 · P5

Say: Read P5. Give time to try before comparing methods.

Ask: Which two quantities are paired? How can you keep the same relationship?

Expect: P5: Yes

If stuck: Put the quantities in two labeled rows. Find one matching pair before scaling.

Slide 94. Watch the thinking | 2 min

Topic: Ratio tables and comparisons
Student: Watch and discuss · worked example

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 15 orange, 10 purple

If stuck: Return to the matching unit in the model.

Slide 95. Your turn | 2 min

Topic: Ratio tables and comparisons
Student: Day 009 · Workbook p. 45 · F

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 6 per part; 24 green; 6 white

If stuck: Compare with the previous example without copying its numbers.

Slide 96. Two ways to see it | 6 min

Topic: Ratio tables and comparisons
Student: Watch and discuss · compare the same example

Say: Partner compare: show the same problem in two ways.

Ask: How does each part of the model match a step in the calculation? Agree or disagree: A mixture with more cups of juice always tastes stronger. Partner compare: Why must the 35-cup paint total be divided into five parts?

Expect: Accept drawings and equations that preserve the same quantities and units. Finds a common water amount or compares part fractions.

If stuck: Accept either method and require a shared comparison unit.

Slide 97. A claim to check | 6 min

Topic: Ratio tables and comparisons
Student: Day 009 · Workbook p. 46 · D

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: There are five parts in all. Yellow is $\frac{2}{5}$ of the paint.

If stuck: Join the two yellow parts and three blue parts into one total strip.

Slide 98. Words and meaning | 3 min

Topic: Ratio tables and comparisons
Student: Watch and discuss

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: (2, 5), (4, 10), and (6, 15). Two cups concentrate matches five cups water. 2 red and 3 blue makes red $\frac{2}{5}$ of all tiles.

If stuck: Read one definition and draw its example.

Slide 99. Mixed review | 10 min

Topic: Ratio tables and comparisons
Student: Day 009 · Workbook p. 47 · M1, M2, M3

Say: Choose a method for each problem.

Ask: What tells you which operation to use?

Expect: Larger; 8; 6 pieces; $\frac{7}{8}$

If stuck: Name the quantities and units before calculating.

Slide 100. In your own words | 2 min

Topic: Ratio tables and comparisons
Student: Day 009 · Workbook p. 43 · S

Say: Use your own words and one example.

Ask: Agree or disagree: A mixture with more cups of juice always tastes stronger.

Expect: A ratio table records equal groups. Comparing mixtures needs a common amount or a common total.

If stuck: Point to the model and name what each part represents.

Slide 101. Exit ticket | 8 min

Topic: Ratio tables and comparisons
Student: Day 009 · Workbook p. 48 · E

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: Mix A

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 102. The week ahead | 5 min

Topic: Ratio tables and comparisons
Student: Days 010-014 · Workbook pp. 53-62

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 103. Tutor reference

Topic: Ratio tables and comparisons
Student: Tutor only · hidden in slide show

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: Mix A

Got it: Chooses A and compares equivalent ratios or correct fractions of juice.

Almost there: Chooses A but does not show a valid comparison, or makes a minor arithmetic slip with the right setup.

Needs re-teaching: Says equal because the differences match, or compares only the juice counts.

PowerPoint answer review

After Friday, outside the weekend lesson time. Say: Check one completed item at a time. Ask: Where does your method first differ from this one? Expect: The answers and worked reasoning printed in the complete keys below. If stuck: Rebuild the quantity with its model, then have the student repair the first incorrect step.

Keep these slides out of view during independent work. The index below follows the answer-review slides in order; every answer is also printed in the complete keys.

Slide 104. Week 2 answers

Next lesson: slide 128. Use this review section after Friday.

Slide 105. Equivalent ratios (1/4)

Items: S03:R1; S03:R2; S03:R3; S03:R4. Follow along: Day 008 · Workbook p. 36 · Items R1, R2, R3, R4

Slide 106. Equivalent ratios (2/4)

Items: S03:S; S03:P1; S03:P2; S03:P3. Follow along: Day 008 · Workbook pp. 36, 37 · Items S, P1, P2, P3

Slide 107. Equivalent ratios (3/4)

Items: S03:P4; S03:P5; S03:P6; S03:F; S03:D. Follow along: Day 008 · Workbook pp. 38, 39 · Items P4, P5, P6, F, D

Slide 108. Equivalent ratios (4/4)

Items: S03:E. Follow along: Day 008 · Workbook p. 40 · Items E

Slide 109. Ratio tables and comparisons (1/4)

Items: S04:R1; S04:R2; S04:R3; S04:R4. Follow along: Day 009 · Workbook p. 43 · Items R1, R2, R3, R4

Slide 110. Ratio tables and comparisons (2/4)

Items: S04:S; S04:P1; S04:P2; S04:P3. Follow along: Day 009 · Workbook pp. 43, 44 · Items S, P1, P2, P3

Slide 111. Ratio tables and comparisons (3/4)

Items: S04:P4; S04:P5; S04:F; S04:D; S04:M1. Follow along: Day 009 · Workbook pp. 45, 46, 47 · Items P4, P5, F, D, M1

Slide 112. Ratio tables and comparisons (4/4)

Items: S04:M2; S04:M3; S04:E. Follow along: Day 009 · Workbook pp. 47, 48 · Items M2, M3, E

Slide 113. Study guide: Try it

Items: W02_Study_Try:1; W02_Study_Try:2; W02_Study_Try:3; W02_Study_Try:4; W02_Study_Try:5. Follow along: Day 009 · Workbook p. 52 · Items 1, 2, 3, 4, 5

Slide 114. Daily practice (1/3)

Items: Day010_Practice:1; Day010_Practice:2; Day010_Practice:3; Day010_Practice:4; Day010_Practice:5. Follow along: Day 010 · Workbook p. 53 · Items 1, 2, 3, 4, 5

Slide 115. Daily practice (2/3)

Items: Day010_Practice:6; Day010_Practice:7; Day010_Practice:8; Day010_Practice:9; Day010_Practice:10. Follow along: Day 010 · Workbook pp. 53, 54 · Items 6, 7, 8, 9, 10

Slide 116. Daily practice (3/3)

Items: Day010_Practice:11; Day010_Practice:12; Day010_Practice:13. Follow along: Day 010 · Workbook p. 54 · Items 11, 12, 13

Slide 117. Daily practice (1/3)

Items: Day011_Practice:1; Day011_Practice:2; Day011_Practice:3; Day011_Practice:4; Day011_Practice:5. Follow along: Day 011 · Workbook p. 55 · Items 1, 2, 3, 4, 5

Slide 118. Daily practice (2/3)

Items: Day011_Practice:6; Day011_Practice:7; Day011_Practice:8; Day011_Practice:9; Day011_Practice:10. Follow along: Day 011 · Workbook pp. 55, 56 · Items 6, 7, 8, 9, 10

Slide 119. Daily practice (3/3)

Items: Day011_Practice:11; Day011_Practice:12; Day011_Practice:13. Follow along: Day 011 · Workbook p. 56 · Items 11, 12, 13

Slide 120. Daily practice (1/3)

Items: Day012_Practice:1; Day012_Practice:2; Day012_Practice:3; Day012_Practice:4; Day012_Practice:5. Follow along: Day 012 · Workbook p. 57 · Items 1, 2, 3, 4, 5

Slide 121. Daily practice (2/3)

Items: Day012_Practice:6; Day012_Practice:7; Day012_Practice:8; Day012_Practice:9; Day012_Practice:10. Follow along: Day 012 · Workbook pp. 57, 58 · Items 6, 7, 8, 9, 10

Slide 122. Daily practice (3/3)

Items: Day012_Practice:11; Day012_Practice:12; Day012_Practice:13. Follow along: Day 012 · Workbook p. 58 · Items 11, 12, 13

Slide 123. Daily practice (1/3)

Items: Day013_Practice:1; Day013_Practice:2; Day013_Practice:3; Day013_Practice:4; Day013_Practice:5. Follow along: Day 013 · Workbook p. 59 · Items 1, 2, 3, 4, 5

Slide 124. Daily practice (2/3)

Items: Day013_Practice:6; Day013_Practice:7; Day013_Practice:8; Day013_Practice:9; Day013_Practice:10. Follow along: Day 013 · Workbook pp. 59, 60 · Items 6, 7, 8, 9, 10

Slide 125. Daily practice (3/3)

Items: Day013_Practice:11; Day013_Practice:12; Day013_Practice:13; Day013_Practice:14. Follow along: Day 013 · Workbook p. 60 · Items 11, 12, 13, 14

Slide 126. Weekly check (1/2)

Items: Day014_Practice:1; Day014_Practice:2; Day014_Practice:3; Day014_Practice:4; Day014_Practice:5. Follow along: Day 014 · Workbook p. 61 · Items 1, 2, 3, 4, 5

Slide 127. Weekly check (2/2)

Items: Day014_Practice:6; Day014_Practice:7; Day014_Practice:8; Day014_Practice:9; Day014_Practice:10. Follow along: Day 014 · Workbook p. 62 · Items 6, 7, 8, 9, 10

Weekday teaching and review

Use one day at a time. Aim for 15-20 minutes, or 15 minutes during a holiday week. Accuracy and a clear explanation matter more than finishing quickly.

Day 010. Monday | pages 53-54

Ask for a model from Saturday before checking an answer. Let the student choose a method. After checking, have the student correct the first step that changed the quantity.

Built-in Booster review: Ratios and rates: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 011. Tuesday | pages 55-56

Connect the practice to Sunday's model. Ask what each number counts and how a multiplication check, diagram, or estimate supports the answer.

Built-in Booster review: Ratios and rates: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 012. Wednesday | pages 57-58

Let the student choose the method for each mixed problem. For the puzzle, ask which conditions are already met. Accept any solution that meets every condition.

Built-in Booster review: Weekly puzzle: items 13. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 013. Thursday | pages 59-60

Ask the student to explain the planted error before repairing it. Optional Stretch comes after the required work and may be skipped without delaying the next day.

Built-in Booster review: Fractions: compare and combine: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 014. Friday | pages 61-62

Give the weekly check independently, without coaching or a calculator. Score the required reasoning as well as the answer. Afterward, choose one error to discuss calmly.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

After Friday

On a 10-item check: 9-10 correct is strong; 7-8 is on track; 5-6 needs another review of the missed topic; 0-4 calls for 20 minutes of re-teaching at the next session. Record the skill and one observation. Check it again with a fresh item at least a week later.

Complete answers for the week

Match the Day number, item ID, and student page before checking. Questions stay in the student workbook; these keys give every answer and the needed reasoning. Equivalent correct forms and sound alternative methods are welcome. An explanation item needs a reason, model, or valid check as well as its conclusion.

Equivalent ratios

Day 008, student workbook pages 36-40

R1 [F-DIV] | student p. 36

Answer: $\frac{1}{6}$ liter

Reasoning: Split the given fractional amount into the stated number of equal shares.

Watch for $\frac{2}{3}$: Multiplying by the divisor answers a different question.

R2 [F-COMP] | student p. 36

Answer: $\frac{5}{8} > \frac{3}{5}$

Reasoning: Rename as twenty-five fortieths and twenty-four fortieths.

Watch for $\frac{5}{8} < \frac{3}{5}$: Compare equal-sized parts, not the original denominator counts.

R3 [F-COMP] | student p. 36

Answer: $\frac{1}{3} > \frac{1}{4}$

Reasoning: With the same numerator, the fraction with larger pieces is larger.

Watch for $\frac{1}{3} < \frac{1}{4}$: A larger denominator makes smaller equal parts, not a larger fraction.

R4 [F-MULT] | student p. 36

Answer: $\frac{3}{5}$

Reasoning: An area model shows 12 of 20 equal parts, or three fifths.

S [EXPLAIN] | student p. 36

Answer: Answers vary. Equivalent ratios multiply or divide both quantities by the same factor.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [R-RATIO] | student p. 37

Answer: 4 : 3; 3 : 4

Reasoning: Name the requested first color before writing its count.

P2 [R-EQUIV] | student p. 37

Answer: 9 nuts

Reasoning: Twelve is three times four, so use three times three nuts.

P3 [R-EQUIV] | student p. 37

Answer: 4 : 3

Reasoning: Make five equal groups. Each has four red and three blue.

P4 [R-EQUIV] | student p. 38

Answer: 6 cups

Reasoning: Nine is 1 $\frac{1}{2}$ times six. Multiply four by the same factor.

P5 [R-EQUIV] | student p. 38

Answer: 6 cups yellow

Reasoning: Fifteen cups blue makes three batches. Each batch uses two cups yellow.

P6 [R-EQUIV] | student p. 38

Answer: 12 builders

Reasoning: Eight testers make four groups of two. Four groups need twelve builders.

F [R-EQUIV] | student p. 39

Answer: 2 batches; 4 cups seeds

Reasoning: Use the worked example in the slides to check each step.

D [R-EQUIV] | student p. 39

Answer: No. Two copies would give 8 : 6, not 7 : 6.

Reasoning: She uses equal additions rather than equal multiplication. Build two full copies of the original group and compare.

E [R-EQUIV] | student p. 40

Answer: 21 green beads

Reasoning: Twelve white beads are three groups of four. Use three groups of seven green.

Watch for 15: Adding eight to each count changes the relationship.

Ratio tables and comparisons

Day 009, student workbook pages 43-48

R1 [F-LINE] | student p. 43

Answer: At 1 $\frac{3}{4}$, the seventh quarter-step from 0

Reasoning: Four fourths reach 1. Three more fourths reach one and three fourths.

R2 [F-SUB] | student p. 43

Answer: $\frac{5}{12}$

Reasoning: Rename both fractions with a common denominator. Subtract equal parts.

R3 [F-SCALE] | student p. 43

Answer: Smaller; 6

Reasoning: Two thirds takes less than all nine.

R4 [F-MULT] | student p. 43

Answer: 2

Reasoning: Count two thirds for each whole-number group, then rename the result.

S [EXPLAIN] | student p. 43

Answer: Answers vary. A ratio table records equal groups. Comparing mixtures needs a common amount or a common total.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [R-TABLE] | student p. 44

Answer: 10 cups water

Reasoning: Double both numbers in (2, 5).

P2 [R-TABLE] | student p. 44

Answer: 25 cups water

Reasoning: Table: multiply (2, 5) by five. Line: five equal jumps of 2 match five jumps of 5.

P3 [R-COMP] | student p. 44

Answer: Mix A

Reasoning: Double A to get 6 syrup : 8 water. Compare 6 syrup with 5 syrup.

P4 [R-WHOLE] | student p. 45

Answer: 14 cups yellow and 21 cups blue

Reasoning: Five equal ratio parts total 35. Each part is 7 cups.

P5 [R-GRAPH] | student p. 45

Answer: Yes

Reasoning: Each y-value is three times its x-value. Zero bags have zero marbles.

F [R-COMP] | student p. 45

Answer: 6 per part; 24 green; 6 white

Reasoning: Use the worked example in the slides to check each step.

D [R-COMP] | student p. 46

Answer: There are five parts in all. Yellow is $\frac{2}{5}$ of the paint.

Reasoning: He uses the other part as the whole. Join the two yellow parts and three blue parts into one total strip.

M1 [F-SCALE] | student p. 47

Answer: Larger; 8

Reasoning: Four thirds takes one whole six plus one third of six.

M2 [F-DIV] | student p. 47

Answer: 6 pieces

Reasoning: Each meter contains 3 equal pieces. Multiply by 2 meters.

Watch for $\frac{2}{3}$: Multiplying by the divisor answers a different question.

M3 [F-COMP] | student p. 47

Answer: $\frac{7}{8}$

Reasoning: The gaps from 1 are one eighth and one fifth. One eighth is smaller.

E [R-COMP] | student p. 48

Answer: Mix A

Reasoning: A is $\frac{4}{7}$ juice; B is $\frac{6}{11}$. Compare $\frac{44}{77}$ and $\frac{42}{77}$.

Watch for They are the same.: Both differences are one, but the fractions of juice differ.

Study guide: Try it

Day 009, student workbook pages 49-52

1 [R-RATIO] | student p. 52

Answer: 5 : 2; 2 : 5

Reasoning: Keep the requested first color first.

2 [R-EQUIV] | student p. 52

Answer: Yes

Reasoning: Multiply both numbers in 6 : 10 by one and one half.

3 [R-TABLE] | student p. 52

Answer: 21

Reasoning: Multiply both entries of the first pair by three.

4 [R-TABLE] | student p. 52

Answer: 32 laps

Reasoning: Four jumps of three minutes match four jumps of eight laps.

5 [R-GRAPH] | student p. 52

Answer: Yes

Reasoning: Every vertical value is twice the horizontal value.

Daily practice

Day 010, student workbook pages 53-54

1 [N-FACT] | student p. 53

Answer: 12

2 [N-FACT] | student p. 53

Answer: 28

3 [N-FACT] | student p. 53

Answer: 16

4 [N-FACT] | student p. 53

Answer: 36

5 [F-COMP] | student p. 53

Answer: $\frac{1}{3} > \frac{1}{4}$

Reasoning: With the same numerator, the fraction with larger pieces is larger.

Watch for $\frac{1}{3} < \frac{1}{4}$: A larger denominator makes smaller equal parts, not a larger fraction.

6 [D-OPS] | student p. 53

Answer: 2.75

7 [N-FACT] | student p. 53

Answer: 25

8 [R-EQUIV] | student p. 53

Answer: 3 cups

Reasoning: Find the number of recipe batches by dividing the rice amount by 3. Each batch needs 2 cups beans.

9 [R-RATIO] | student p. 54 | Built-in review: Ratios and rates

Answer: 5 : 2; 2 : 5

Reasoning: Keep the requested first color first.

10 [R-EQUIV] | student p. 54 | Built-in review: Ratios and rates

Answer: Yes

Reasoning: Multiply both numbers in 6 : 10 by one and one half.

11 [R-EQUIV] | student p. 54 | Built-in review: Ratios and rates

Answer: No

Reasoning: Adding two to each count changes the relationship. Two batches would give 6 : 14.

Watch for Yes: Equal additions do not preserve a ratio.

12 [R-EQUIV] | student p. 54 | Built-in review: Ratios and rates

Answer: 24 runners

Reasoning: Eighteen walkers make six groups. Six groups need twenty-four runners.

13 [R-EQUIV] | student p. 54

Answer: No. Two copies would give 8 : 6, not 7 : 6.

Watch for 7 : 6 is equivalent to 4 : 3.: She uses equal additions rather than equal multiplication.

Daily practice

Day 011, student workbook pages 55-56

1 [F-EQUIV] | student p. 55

Answer: 0.5; 50%

2 [F-EQUIV] | student p. 55

Answer: 0.25; 25%

3 [F-EQUIV] | student p. 55

Answer: 0.75; 75%

4 [F-EQUIV] | student p. 55

Answer: 0.125; 12.5%

5 [F-COMP] | student p. 55

Answer: $\frac{1}{3} > \frac{1}{4}$

Reasoning: With the same numerator, the fraction with larger pieces is larger.

Watch for $\frac{1}{3} < \frac{1}{4}$: A larger denominator makes smaller equal parts, not a larger fraction.

6 [N-SCALE] | student p. 55

Answer: 3

7 [N-FACT] | student p. 55

Answer: 12

8 [R-WHOLE] | student p. 55

Answer: 8 yellow; 20 purple

Reasoning: Add ratio parts: 2 + 5 = 7. Divide the total by 7, then multiply each ratio part.

9 [R-TABLE] | student p. 56 | Built-in review: Ratios and rates

Answer: 21

Reasoning: Multiply both entries of the first pair by three.

10 [R-TABLE] | student p. 56 | Built-in review: Ratios and rates

Answer: 32 laps

Reasoning: Four jumps of three minutes match four jumps of eight laps.

11 [R-GRAPH] | student p. 56 | Built-in review: Ratios and rates

Answer: Yes

Reasoning: Every vertical value is twice the horizontal value.

12 [R-WHOLE] | student p. 56 | Built-in review: Ratios and rates

Answer: $\frac{3}{8}$

Reasoning: Count all eight cups before forming the fraction.

Watch for $\frac{3}{5}$: Five counts the other part, not the whole.

13 [R-COMP] | student p. 56

Answer: There are five parts in all. Yellow is $\frac{2}{5}$ of the paint.

Watch for $\frac{2}{3}$: He uses the other part as the whole.

Daily practice

Day 012, student workbook pages 57-58

1 [N-SCALE] | student p. 57

Answer: 42

2 [N-SCALE] | student p. 57

Answer: 4

3 [N-SCALE] | student p. 57

Answer: 9

4 [N-SCALE] | student p. 57

Answer: 12

5 [R-EQUIV] | student p. 57

Answer: Yes

Reasoning: Multiply both numbers in 6 : 10 by one and one half.

6 [R-EQUIV] | student p. 57

Answer: No

Reasoning: Adding two to each count changes the relationship. Two batches would give 6 : 14.

Watch for Yes: Equal additions do not preserve a ratio.

7 [F-MULT] | student p. 57

Answer: 2

Reasoning: Count two thirds for each whole-number group, then rename the result.

8 [F-ADD] | student p. 57

Answer: $\frac{5}{6}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

9 [D-OPS] | student p. 58

Answer: 4.75

10 [G-VOL] | student p. 58

Answer: 36 cm³

11 [N-FACT] | student p. 58

Answer: 14

12 [G-VOL] | student p. 58

Answer: 42 cm³

13 [PUZZLE] | student p. 58 | Built-in review: Weekly puzzle

Answer: 15 red and 10 blue

Reasoning: One batch contains five tiles. The only multiple of five strictly between 20 and 30 is 25.

Daily practice

Day 013, student workbook pages 59-60

1 [N-EST] | student p. 59

Answer: 400

2 [N-EST] | student p. 59

Answer: 5

3 [N-EST] | student p. 59

Answer: About 100

4 [N-EST] | student p. 59

Answer: About 20

5 [R-EQUIV] | student p. 59

Answer: Yes

Reasoning: Multiply both numbers in 6 : 10 by one and one half.

6 [R-EQUIV] | student p. 59

Answer: No

Reasoning: Adding two to each count changes the relationship. Two batches would give 6 : 14.

Watch for Yes: Equal additions do not preserve a ratio.

7 [R-EQUIV] | student p. 59

Answer: 4 : 7

Reasoning: Divide both counts by the same factor.

8 [F-DIV] | student p. 59

Answer: $\frac{1}{18}$; stories vary

Reasoning: Example: share one sixth liter equally among three cups. Each gets one eighteenth liter.

9 [F-ADD] | student p. 60 | Built-in review: Fractions: compare and combine

Answer: $\frac{5}{6}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

10 [F-ADD] | student p. 60 | Built-in review: Fractions: compare and combine

Answer: $\frac{11}{12}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

11 [F-ADD] | student p. 60 | Built-in review: Fractions: compare and combine

Answer: $\frac{7}{12}$

12 [F-ADD] | student p. 60 | Built-in review: Fractions: compare and combine

Answer: $\frac{3}{4}$

13 [R-EQUIV] | student p. 60

Answer: 24 runners

Reasoning: Eighteen walkers make six groups. Six groups need twenty-four runners.

14 [R-COMP] | student p. 60

Answer: There are five parts in all. Yellow is $\frac{2}{5}$ of the paint.

Watch for $\frac{2}{3}$: He uses the other part as the whole.

Weekly check

Day 014, student workbook pages 61-62

1 [R-EQUIV] | student p. 61

Answer: Yes

Reasoning: Multiply both numbers in 6 : 10 by one and one half.

2 [R-EQUIV] | student p. 61

Answer: No

Reasoning: Adding two to each count changes the relationship. Two batches would give 6 : 14.

Watch for Yes: Equal additions do not preserve a ratio.

3 [R-EQUIV] | student p. 61

Answer: 4 : 7

Reasoning: Divide both counts by the same factor.

4 [R-EQUIV] | student p. 61

Answer: 4 : 7

Reasoning: Divide both counts by the same factor.

5 [F-COMP] | student p. 61

Answer: $\frac{2}{4} > \frac{2}{5}$

Reasoning: With the same numerator, the fraction with larger pieces is larger.

Watch for $\frac{2}{4} < \frac{2}{5}$: A larger denominator makes smaller equal parts, not a larger fraction.

6 [F-LINE] | student p. 62

Answer: At 1 $\frac{3}{4}$, the seventh quarter-step from 0

Reasoning: Four fourths reach 1. Three more fourths reach one and three fourths.

7 [F-MULT] | student p. 62

Answer: $\frac{3}{5}$

Reasoning: An area model shows 12 of 20 equal parts, or three fifths.

8 [D-OPS] | student p. 62

Answer: 6.75

9 [G-VOL] | student p. 62

Answer: 90 cm^3

10 [R-EQUIV] | student p. 62

Answer: No. Two copies would give 8 : 6, not 7 : 6.

Watch for 7 : 6 is equivalent to 4 : 3.: She uses equal additions rather than equal multiplication.